



# Seemingly same but actually different: A comparative study of class friendship networks and support-seeking networks

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## Abstract

Peer support in school is vital for students' mental health, particularly within China's distinctive educational context centered on stable, fixed classes. **Class friendship networks and support-seeking networks are widely studied interpersonal networks in peer support research. Previous studies have rarely conducted in-depth comparisons between these two types of networks, leading to confusion in their usage,** which impedes the further development of school peer support research. This **cross-sectional study** examines a sample of 799 second-year junior high school students across 29 classes (48.6% female) from **China**, exploring the **connections and differences between the two networks by comparing their structures and functions through social network analysis.** The results indicate the following: **(1) The two networks exhibit partial overlap,** though there are differences in structural indicators at individual, dyadic and triadic, and group level; **(2) While the two networks share similar key configurations, the role of these configurations in network formation differs;** **(3) Perceived peer support exerts a greater influence on the formation of support-seeking networks;** **(4) Both networks are closely associated with mental health, but support-seeking networks show stronger associations with well-being than friendship networks.** The findings suggest that although the two networks are somewhat interconnected, they differ in structure, formation processes, and their impact on students, indicating they are distinct types of networks within class-based peer environments. Therefore, **they should be differentiated in future research** and applications.

**Keywords** Class peer support · Exponential random graph models · Friendship networks · Support-seeking networks · Social network analysis

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## Introduction

Peer support refers to the social support provided by peers and plays a significant role in the development of primary and middle school students (Letkiewicz et al., 2023). Peer groups constitute a primary social environment for students' interaction, with support within these groups influencing students at various levels, such as peer support at the individual level and peer climate at the group level (Letkiewicz et al., 2023; Wang et al., 2020). For students, for whom school constitutes the primary environment for socialization, investigating the nature and function of multi-level peer support within the school context is of paramount importance. In the Chinese context, where a fixed-class system is implemented in most primary and secondary schools, students are grouped into stable classes for most academic and social activities. This system operates not only as an instructional unit but also as the core social-organizational unit of students' daily lives. Students typically remain in the same class, with the same peers and a fixed head teacher, throughout several years, engaging together in virtually all academic and extracurricular activities. This high-intensity, long-term coexistence makes peer interactions within the class exceptionally frequent and deep. Furthermore, rooted in a collectivist cultural tradition that emphasizes belonging and responsibility to the group, the Chinese "class collective" is imbued with strong emotional and normative significance, deeply embedding students within a relationship network where peer influence is particularly potent. Hence, the profound and enduring relationships that develop among classmates render an analysis of these levels of peer support within this structure particularly valuable. Social network analysis offers a crucial framework for exploring the characteristics and effects of class peer support at various levels (Lee et al., 2005; Li & Stone, 2018).

Class friendship networks (FNs) and support-seeking networks (SSNs) are two common types of peer support networks in schools. FNs consist of all members within a class and the friendship relationships among them (Li & Stone, 2018). These networks represent the broad structure of affective and social ties, through which students exchange viewpoints, emotions, and resources, thereby deriving peer support. SSNs consist of all members and their support-seeking relationships. They are defined by specific, instrumental interactions where students actively seek help to address challenges. While both facilitate peer support, they stem from fundamentally different relational dynamics: FNs constitute the pervasive backdrop of social affiliation, whereas SSNs capture deliberate acts of assistance-seeking. Research confirms the benefits of support-seeking, linking it to better academic performance, mental health, and well-being (Barrow & Thomas, 2022; Li et al., 2023; Melkman,

2017). Since both networks provide peer support, some researchers collectively refer to them as peer support networks (Li & Stone, 2018; Wang et al., 2021). However, this broad categorization obscures their distinct foundations. These foundational differences likely give rise to divergent network structures, are shaped by different factors, and are ultimately associated with distinct aspects of student mental health, such as in their respective contributions to enhancing overall well-being versus alleviating specific internalizing problems. Consequently, failing to distinguish between them may obscure their differences and hinder both research precision and network-specific applications.

In conclusion, distinguishing the structure, formation processes, and impact on students of these two types of class networks is vital for differentiating relevant research findings. This distinction is essential for selecting the appropriate network in future research and applications, and it is also important for promoting the healthy development of students by promoting the development of FNs or SSNs.

## Literature

### The basic foundational elements of the two types of peer support networks

The connection and differences between class FNs and SSNs are first reflected in their fundamental components. Friendship relationships and support-seeking relationships serve as the foundational elements of these networks respectively (Li & Stone, 2018; Wang et al., 2021). Comparing these two types of relationships is essential for understanding the connections and differences between the networks.

Friendship relationships provide various forms of support. Firstly, students can seek and receive direct assistance from friends to address daily issues (Zander et al., 2019). Secondly, friendship relationships may offer subtle, implicit forms of support that are less overt, such as a sense of belonging and security (Santini et al., 2019; Wu & Sheng, 2019). However, friendship relationships can also have negative consequences. For instance, the support actively provided by friends may exceed others' needs, potentially undermining mental well-being (Rubin et al., 2011). In summary, friendship relationships provide complex support, yet certain aspects can be detrimental.

The role of support-seeking relationships can be analyzed through the support-seeking process: individuals identify their problems, evaluate potential supporters (based on factors such as closeness and support capabilities), activate support-seeking relationships, and, following communication, receive assistance (Barbee & Cunningham, 1995). In the school setting, students often encounter challenges in

academics and interpersonal relationships (Wan & Meng, 2024). Actively seeking support is crucial for addressing these challenges and fostering mental well-being of students (Letkiewicz et al., 2023).

Friendship relationships and support-seeking relationships overlap to some degree, yet they also exhibit key differences. On one hand, students tend to seek help from friends with the ability to assist (Barbee & Cunningham, 1995), which may result in some overlap between the two relationships. Research has shown that students seek academic help from friends (Zander et al., 2019), and turn to friends with relevant knowledge when dealing with emotional problems (van den Toren et al., 2020). On the other hand, the motivations for forming friendship and support-seeking relationships differ, which may lead to individual variances between the two relationships, thereby creating structural distinctions between the networks. For example, students might seek academic support from classmates outside their friend circle who excel in mathematics (Zander et al., 2019), or may seek help with emotional issues from non-friends with specific expertise, such as experience in intimate relationships or emotional communication (van den Toren et al., 2020). Based on these observations, this study hypothesizes that the two networks overlap partially, with the basic relationships in the same class partially overlapping (H1).

### Comparing the structural indicators and configurations between the two networks

FNs and SSNs may display structural differences across multiple levels. Establishing support-seeking relationships typically requires more complex conditions than friendship relationships (Barbee & Cunningham, 1995), and these differences may manifest in structural indicators at various levels within the networks. At the individual level, out-degree and in-degree centrality indicate the number of relationships an individual initiates and receives (Liu, 2019). In this study, the number of relationships in FNs is referred to as “friendship out-degree” and “friendship in-degree”, while in SSNs, these are termed “support out-degree” and “support in-degree”. For example, if a student is nominated as a friend by  $n$  classmates, their friendship in-degree is  $n$ ; if the student nominates  $m$  classmates as friends, their friendship out-degree is  $m$ . Similarly, support in-degree refers to the number of classmates who seek support from the student ( $n$ ), while support out-degree refers to the number of classmates from whom the student seeks support ( $m$ ). Because support-seeking relationships are harder to establish, the out-degree and in-degree in FNs are likely to exceed the corresponding values in SSNs. At the dyadic and triadic levels, the status of members’ relationships reflects the local

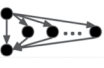
characteristics of the network. For instance, relationships between two members can be classified as reciprocal, non-reciprocal, or nonexistent (Liu, 2019), and there are sixteen possible relationship combinations among three members (Butts, 2024). Given the ease of relationship formation, the number of connected dyads and triads in FNs should be greater than in SSNs. Finally, group-level indicators capture the overall structural characteristics of the network, such as network density, which is the ratio of the actual number of relationships to the total possible relationships, indicating the overall closeness of the network (Liu, 2019). As friendship relationships are easier to form, the density of FNs within a class is expected to be higher. In conclusion, given that the differences between friendship and support-seeking relationships can influence all network levels, we hypothesize that significant differences in structural indicators exist between the two networks (H2).

The structural similarities and differences between the two networks may also arise from the gradual accumulation of local structures during the network-building process. By analyzing the characteristics of the two networks in terms of local configurations using Exponential Random Graph Models (ERGM; Lusher et al., 2013), we can explore and compare the local features of the two networks and investigate the reasons for their structural similarities and differences. Table 1 presents common local configurations in ERGMs. Previous research indicates that positive relationship networks typically exhibit similar local configuration patterns (Lusher et al., 2013). As both FNs and SSNs are positive relationship networks, they may share similar local configurations. However, the importance of these configurations may differ between the two networks. For example, support-seeking relationships require consideration of problem-solving abilities and are generally harder to establish, making them fewer in number compared to friendship relationships within the same class (Wang et al., 2021). This may be reflected in the local configurations, with Activity playing a more significant role in FNs (van Rijsewijk et al., 2020). Therefore, we hypothesize that while the two networks share similar local configuration categories, the roles of these configurations differ between them (H3).

### Comparing the impact of perceived social support on the two networks

Social support assists individuals in managing interpersonal challenges, building mutual trust, providing stability and intimacy, and promoting the formation and maintenance of relationships (Barbee & Cunningham, 1995; Virtanen & Isotalus, 2011). Both FNs and SSNs may be influenced by class members’ perceptions of social support, although this influence may vary between the two networks, with

**Table 1** Local configurations and effects of directed networks in ERGMs

| Name                          | Symbols                                                                           | Meaning of Configurations or Effects                                    | The Meaning of Configurations and Effects in This Study                                                               |
|-------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Local Configuration</b>    |                                                                                   |                                                                         |                                                                                                                       |
| <i>Arc</i>                    |  | The baseline tendency of relationship emergence.                        | The baseline tendency for relationship emergence in FNs or SSNs.                                                      |
| <i>Reciprocity</i>            |  | Tendency for mutual selection.                                          | Two students select each other as friends (or supporters).                                                            |
| <i>Popularity</i>             |  | Tendency for in-degree centrality to be concentrated on certain nodes.  | In a class, some students are chosen by more classmates as friends (or supporters).                                   |
| <i>Activity</i>               |  | Tendency for out-degree centrality to be concentrated on certain nodes. | In a class, some students choose more classmates as friends (or supporters)                                           |
| <i>Triadic transitivity</i>   |  | Transitivity of triadic relationships.                                  | When student i chooses h, and h chooses j as a friend (or supporter), i tends to choose j as a friend (or supporter). |
| <i>Path closure</i>           |  | Path closure is more concentrated in certain areas of the network.      | Tendency for triadic transitivity to occur in clusters based on the relationship between two students.                |
| <i>Cyclic closure</i>         |  | Triadic cycles are more concentrated in certain areas of the network.   | Tendency for triadic cyclicity to occur in clusters based on the relationship between two students.                   |
| <b>Actor relation effects</b> |                                                                                   |                                                                         |                                                                                                                       |
| <i>Sender Effect</i>          |  | People with higher scores on feature X initiate more relationships.     | Students with high perceived peer support tend to choose more friends (or supporters).                                |
| <i>Receiver Effect</i>        |  | People with higher scores on feature X receive more relationships.      | Students with high perceived peer support tend to be chosen by more people as friends (or supporters).                |

1) Each black dot in the table represents an actor, that is, a student in the class; directed lines indicate a directed relationship from the sender to the receiver; 2) The parameter *Arc* is not a direct measurement of network density

the impact on SSNs potentially being stronger. Perceived social support is strongly correlated with students' attitudes and behaviors toward seeking help; those who perceive higher levels of social support tend to believe that others are willing to help them, adopt a more positive attitude toward seeking assistance, and are more likely to seek support (Gao et al., 2020). As a result, students who perceive higher levels of peer support are more inclined to establish support-seeking relationships, thus facilitating the development of SSNs. In contrast, friendship relationships, which are strong emotional bonds between two individuals (Rubin et al., 2011), are relatively less influenced by perceived peer support. Therefore, the impact of perceived peer support on FNs may be smaller. Based on these considerations, this study hypothesizes that perceived peer support has a greater impact on SSNs than on FNs (H4).

### Comparison of the impact of friendship networks and support-seeking networks on students' mental health

Friendship is a key task for the social development of primary and secondary school students and plays a significant role in their mental health. Friendships provide companionship, intimacy, encouragement, and empathy, fulfilling students' emotional needs and promoting their well-being (Alsarrani et al., 2022; Lei, 2021; Niu et al., 2021). Moreover, they help reduce internalizing problems such as depression, anxiety, and loneliness (He & Wang, 2023; Yang et al., 2020). In the class context, having more friends

increases opportunities to obtain companionship, intimacy, and empathy. Consequently, this study hypothesizes that students with higher friendship out-degree and in-degree will experience greater well-being and fewer internalizing problems (H5.1).

SSNs also have a close relationship with students' mental health. In the class, students use peer support to resolve confusion, express emotions, relieve stress, and address academic and interpersonal challenges. This support contributes to increased well-being, improved academic performance, and reduced internalized problems (Cheng & Huang, 2021; Letkiewicz et al., 2023; Webster et al., 2021). Students with more supportive relationships and access to potential support providers are more likely to receive appropriate assistance. Being sought for support also boosts students' sense of life meaning and self-worth, enhancing their well-being (Li et al., 2018). Furthermore, being approached for support by more classmates often implies more interpersonal interactions, which helps alleviate loneliness. Based on these considerations, this study hypothesizes that students with higher support out-degree and support in-degree will report greater well-being and fewer internalizing problems (H5.2).

While both networks influence mental health, there may be differences in their impact. FNs primarily address needs for companionship and intimacy. Compared to support relationships, this close emotional connection may be more effective in alleviating emotional issues such as loneliness and depression (Schwartz-Mette et al., 2020). In contrast,

seeking support enables students to cope with problems, enhances problem-solving abilities, and provides a sense of accomplishment in resolving difficulties (Fullerton et al., 2021). This process may go beyond the role of FNs and potentially contribute more significantly to improving positive emotions and life satisfaction. Furthermore, excessive support from friendships may be detrimental to well-being. Therefore, we hypothesize that, when comparing the two types of networks, FNs have a stronger mitigating effect on students' internalizing problems, while SSNs more effectively enhance their well-being (H6).

In sum, this study will explore the connections and distinctions between Friendship Networks (FNs) and Support-Seeking Networks (SSNs). This issue will be examined in a layered and in-depth manner—from the structural overlap and differences between the two (H1), to the connections and variations in structural indicators (H2), to the network configurations and the influence of perceived peer support on both networks (H3 and H4), and finally to the commonalities and differences in how the two networks affect students' mental health (H5 and H6).

## Methods

### Participants

The data for this study were derived from the Longitudinal Study of the Social Development of Children and Adolescents. Although the overarching study is longitudinal, the current analysis uses cross-sectional data from the 2016 assessment wave. This specific wave was selected because it was the only one that included the complete set of key variables required for the present research questions.

In this study, a total of 826 eighth-grade students from 30 classes across six junior high schools in Beijing were selected using cluster sampling in this cross-sectional study, among which two schools were in urban districts and four in suburban areas, thus enhancing the representativeness of the sample. During the analysis phase, students who did not meet the following criteria were excluded: (1) students who did not answer questions seriously (e.g., nominating all students, including themselves), and (2) classes with too few students, as the ERGMs would struggle to obtain reliable parameter estimates. As a result, a class with only four students was excluded. The final sample consisted of 799 students from 29 classes (48.6% female,  $M=14.11$  years old,  $SD=0.48$ ), with class sizes ranging from 20 to 34 students ( $M=27.55$ ,  $SD=4.12$ ). Two classes did not provide responses to questions on perceived peer support and were therefore excluded from the analysis of perceived peer support.

### Procedures

The survey was administered during class hours, and all students in the class participated. A researcher was present in each class to provide instructions and address students' questions as necessary. All researchers were trained undergraduate or graduate students majoring in psychology or education. Students completed a paper-and-pencil questionnaire. Prior to participation, students were informed of the study's purpose and requirements. If they agreed to participate, they signed an informed consent form and were made aware that they could withdraw from the study at any time. The study was conducted in accordance with APA ethical standards and approved by Scientific Research Ethics Committee of the corresponding author's institution (protocol: 2021–36).

### Measures

The broader longitudinal database contains numerous additional constructs (e.g., externalizing behaviors, psychological traits, family dynamics). For the present study, which aims to specifically compare the structures and functions of friendship and support-seeking networks, the analysis was deliberately focused on the network variables, perceived peer support, well-being, and internalizing mental health outcomes. This focused selection was made a priori to maintain theoretical coherence and avoid over-extending the analytical model.

#### Friendship networks and support-seeking networks

**Friendship networks** were assessed using the peer nomination question, "Do you have any good friends in your class? Who are they?" A list of all students in the class was distributed, and students were asked to provide the IDs of their nominated friends.

**Support-seeking networks** were assessed using the peer nomination question, "When you have worries or encounter problems, would you seek help from your classmates? If so, please write down their student IDs" (Wang et al., 2021). Students listed the IDs of classmates from whom they would seek help.

#### Perceived peer support

The friend support dimension of the **Perceived Social Support Scale** (Zimet et al., 1988) was used to measure perceived peer support. This dimension includes four items such as "I can count on my friends when things go wrong." Items were rated on a 5-point scale (0="Strongly disagree" to 4="Strongly agree"), with higher scores indicating

stronger perceptions of peer support. In this study, the internal consistency for the peer support dimension was excellent (Cronbach's  $\alpha=0.893$ ).

### Mental health indicators

**Well-being** was measured using the **Index of Well-Being** developed by Campbell et al. (1976), which consists of nine items divided into two dimensions: the index of general affect and life satisfaction. The index of general affect measures emotional well-being with eight items scored on a 7-point scale. Life satisfaction is measured with a single item, where respondents rate their overall satisfaction from 1 to 7. In this study, the internal consistency for the index of general affect and the overall Index of Well-Being was excellent (Cronbach's  $\alpha=0.933$ , 0.936).

**Anxiety** was measured using the **Chinese version** of the **Children's Manifest Anxiety Scale** (Dong & Lin, 2011; Reynolds & Richmond, 1978). The scale includes 28 items across three dimensions: physiological anxiety, worry/oversensitivity, and social concerns/concentration. Students responded with "yes" (scored as 1) or "no" (scored as 0). Higher scores indicate more severe anxiety. Internal consistency for the three dimensions and overall scale was satisfactory (Cronbach's  $\alpha=0.786$ , 0.847, 0.710, and 0.914, respectively).

**Depression** was assessed using the **Chinese version** of the **Children's Depression Inventory** (Dong & Lin, 2011; Kovacs & Beck, 1977), which contains 10 items. Participants rated their experiences over the past two weeks on a three-point scale (0="I am occasionally unhappy," 1="I am often unhappy," 2="I am always unhappy"). Higher scores indicate more severe depression. The internal consistency for the depression scale was good (Cronbach's  $\alpha=0.831$ ).

**Loneliness** was measured using the **Chinese version** of the **Children's Loneliness Questionnaire** (Asher et al., 1984; Dong & Lin, 2011), which includes 16 items such as "I can easily make new friends at school." The scale uses a 4-point Likert scale, with higher scores indicating greater loneliness. The internal consistency for the loneliness scale was excellent (Cronbach's  $\alpha=0.915$ ).

### Controlled variables

To account for potential confounding factors, the study controlled for gender and family socioeconomic status (SES) when examining the relationships between FNs/SSNs and mental health. Family SES was assessed using a method from the Organization for Economic Co-operation and Development (OECD, 2012), based on three indicators: parents' education level, parents' occupation, and family possessions.

## Analytic strategy

Firstly, we explored the overlap between the two networks. Using Excel functions, we analyzed the relationship status of all dyads within each class (i.e., whether two students were connected in both networks). Specifically, we first constructed adjacency matrices for both the FN and SSN in each class, where "1" indicated the presence of a tie and "0" its absence. The overlap between the two networks was then computed by multiplying the corresponding cells of the two adjacency matrices using Excel's multiplication operator, yielding a new matrix in which "1" represented an overlapping tie and "0" otherwise. The total number of overlapping ties in each class was obtained using the COUNTIF function. Finally, the proportion of overlapping relationships was computed within the total relationships of the SSNs and FNs, respectively. This resulted in the proportion of overlapping relationships in each network for each class. Besides, the QAP (Quadratic Assignment Procedure) correlation coefficients between the networks were calculated using Ucinet 6.0 (Liu, 2019). The results from the steps above can address Hypothesis 1.

Secondly, we examined the relationship between the two networks by calculating structural indicators for all classes at each level using the R packages "sna" and "igraph" (Butts, 2024; Csárdi, 2024). Paired-sample *t*-tests were performed using SPSS 20.0 to compare the differences in indicators between the two networks and to calculate the correlation coefficients for each indicator. As the results in this section involve multiple comparisons, we applied the Benjamini-Hochberg procedure (False Discovery Rate, FDR; Benjamini & Hochberg, 1995) to correct for them. In this study, *q* indicates the corrected *p*-value (FDR-adjusted). The analyses described are suited to examine H2.

Next, MPNet software was employed to explore local configurations in the two networks and the role of perceived peer support (Lusher et al., 2013). The analysis proceeded in three steps: Firstly, a trial-and-error approach identified suitable local configurations for both networks. Secondly, the effects of these configurations were examined. Thirdly, the "Actor-relation effects" in ERGMs were used to analyze the role of perceived peer support in network formation (Lusher et al., 2013). Additionally, SPSS 20.0 was used to compare the effect sizes of various local configurations across both networks. As these comparisons also involve multiple comparisons, we applied FDR to correct for them.

ERGM analysis was conducted in two stages: (1) The local configurations of each network within each class were calculated separately, using consistent model settings for networks within the same category. (2) A meta-analysis was performed to integrate the results across all classes for each type of network, using Mlwin software (Lubbers & Snijders,

2007). Previous studies have identified key local configurations for ERGM analysis, including *Arc*, *Reciprocity*, *Popularity*, *Activity*, *Transitivity*, *Path Closure*, and *Cyclic Closure* (Lusher et al., 2013). Convergence and fitness tests showed that *Arc*, *Reciprocity*, *Popularity*, and *Activity* were suitable for most classes, and these configurations were selected for the analysis. The Actor-relation effects were analyzed using the *Receiver effect* and *Sender effect* of perceived peer support, as detailed in Table 1. These procedures are designed to test H3 and H4.

To evaluate the goodness-of-fit (GOF) of the ERGM specifications, we examined several statistics. Specifically, we assessed the *t*-ratios for each model parameter, the Mahalanobis distance, and the maximum quasi-autocorrelation in absolute value. The *t*-ratio compares observed network statistics with the mean statistics from networks simulated by the fitted model, where values close to zero indicate good fit. In addition, the Mahalanobis distance was used to evaluate the multivariate discrepancy between observed and simulated networks, while the maximum quasi-autocorrelation was examined to assess the mixing of the Markov Chain Monte Carlo (MCMC) simulation and the stability of parameter estimation.

Lastly, Mplus 8.0 was used to assess the relationships between FNs/SSNs and mental health and to compare their effects. Path models were used to examine associations between network centrality (out-degree and in-degree) and mental health indicators, controlling for gender and family SES. The path model was estimated using covariance-based structural equation modeling with the robust maximum likelihood (MLR) estimator. It included four independent variables, five dependent variables, and two control variables, allowing coefficients to be compared within a unified modeling framework. Since the path model involves multiple path coefficients, the correction was also applied to these coefficients by FDR. After that, the Wald test was employed to compare the effects of friendship out-degree versus support out-degree, as well as the effects of friendship in-degree versus support in-degree on mental health. Likewise, the results of the Wald test were also corrected using FDR. The described approach allows for an examination of H5 and H6.

The ERGM and path model serve complementary purposes in addressing our research questions. The ERGM is employed to identify the endogenous structural processes (e.g., reciprocity, popularity) and exogenous actor-relation effects (e.g., the role of perceived peer support) that govern the formation of friendship and support-seeking networks. In contrast, the path model is used to examine the differential predictive associations between key structural positions (i.e., in-degree and out-degree centrality) within these networks and the five psychosocial outcomes. While the

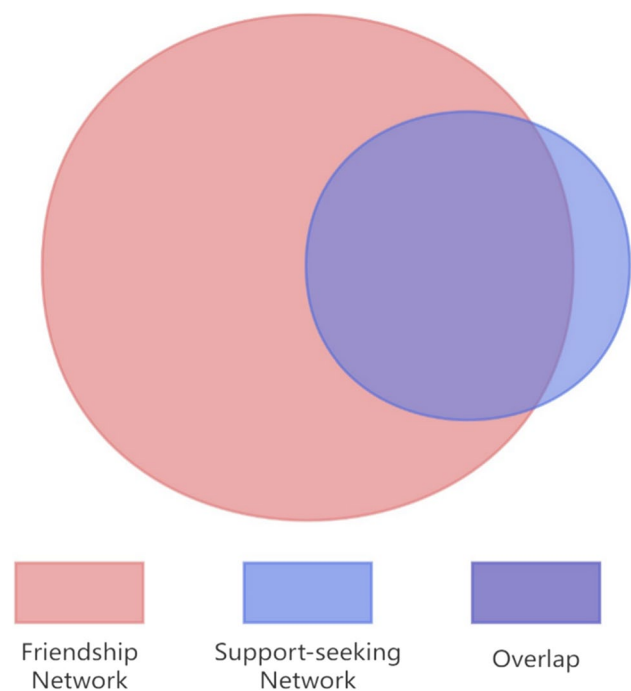
ERGM explains how these networks are structured, the path model assesses how occupying different positions within these networks affects students' psychosocial outcomes. Together, these analyses provide a comprehensive account of the antecedents, structure, and outcomes of adolescent peer networks.

## Results

### Comparison of structural indicators between the two types of networks

The analysis of the overlapping relationships between the two networks (as shown in Fig. 1) reveals that, on average, 28.5% of relationships in FNs overlap, while 88.5% of support relationships in SSNs overlap (min=15%, max=47%,  $SD=7.16\%$  for FNs; min=75%, max=100%,  $SD=6.05\%$  for SSNs). The QAP correlation results demonstrate a moderate correlation between the two networks across all classes ( $r_{\min}=0.34$ ,  $r_{\max}=0.57$ ,  $M=0.45$ ,  $SD=0.06$ ), with all QAP correlations being statistically significant across classes. The results are in line with H1.

Table 2 provides a detailed comparison of the structural indicators for the two types of networks. According to the



**Fig. 1** The relationship between friendship network and support-seeking network. Note: The large circle represents the friendship relationships within a class, while the small circle represents the support-seeking relationships within the same class. The area of the circles indicates the number of relationships in each network, and the overlapping parts represent the number of overlapping relationships

**Table 2** Network structure indicators of 29 classes

| Structure indicators                     | FNs         |            | SSNs        |            | Correlation coefficient | Difference tests | <i>p</i> | <i>q</i> | Cohen's <i>d</i> |
|------------------------------------------|-------------|------------|-------------|------------|-------------------------|------------------|----------|----------|------------------|
|                                          | <i>M</i>    | <i>SD</i>  | <i>M</i>    | <i>SD</i>  |                         |                  |          |          |                  |
| Individual-level indicators              |             |            |             |            |                         |                  |          |          |                  |
| In-degree <sup>a</sup>                   | 5.52        | 2.79       | 1.84        | 1.62       | 0.57**                  | 3.68***          | 0.000000 | 0.000000 | 1.61             |
| Out degree <sup>b</sup>                  | 5.52        | 4.05       | 1.84        | 2.03       | 0.39**                  | 3.68***          | 0.000000 | 0.000000 | 1.15             |
| Dyadic indicators                        |             |            |             |            |                         |                  |          |          |                  |
| Non-reciprocal relationship <sup>c</sup> | 68.10       | 21.41      | 31.83       | 12.36      | 0.56**                  | 36.28***         | 0.000000 | 0.000000 | 2.08             |
| Reciprocal relationship <sup>d</sup>     | 41.83       | 13.18      | 9.34        | 4.07       | 0.44*                   | 32.48***         | 0.000000 | 0.000000 | 3.33             |
| Nonexistent <sup>e</sup>                 | 264.34      | 99.86      | 333.10      | 107.69     | 0.98***                 | −68.76***        | 0.000000 | 0.000000 | −0.66            |
| Same gender <sup>f</sup>                 | 0.84        | 0.21       | 0.88        | 0.29       | 0.66**                  | −0.05***         | 0.000000 | 0.000000 | −0.16            |
| Boy → boy                                | 0.45        | 0.10       | 0.40        | 0.14       | 0.76***                 | 0.05**           | 0.005448 | 0.005837 | 0.43             |
| Boy → girl                               | 0.08        | 0.04       | 0.07        | 0.05       | 0.67***                 | 0.01             | 0.076657 | 0.076657 | 0.28             |
| Girl → girl                              | 0.39        | 0.11       | 0.48        | 0.15       | 0.75***                 | −0.10***         | 0.000007 | 0.000008 | −0.79            |
| Girl → boy                               | 0.08        | 0.05       | 0.05        | 0.04       | 0.60***                 | 0.04****         | 0.000063 | 0.000070 | 0.79             |
| Triadic indicators <sup>g</sup>          |             |            |             |            |                         |                  |          |          |                  |
| A, B, C                                  | 1270.62     | 877.59     | 2400.34     | 1212.55    | 0.93***                 | −1129.72***      | 0.000000 | 0.000000 | −1.07            |
| A->B, C                                  | 859.45      | 394.75     | 632.17      | 290.33     | 0.78***                 | 227.28***        | 0.000035 | 0.000040 | 0.66             |
| A<—>B, C                                 | 577.48      | 257.21     | 188.69      | 93.76      | 0.65***                 | 388.79***        | 0.000000 | 0.000000 | 2.01             |
| A<-B->C                                  | 110.97      | 75.27      | 31.38       | 35.56      | 0.28                    | 79.59***         | 0.000003 | 0.000004 | 1.35             |
| A->B<-C                                  | 47.38       | 31.08      | 15.31       | 11.15      | 0.54**                  | 32.07***         | 0.000001 | 0.000001 | 1.37             |
| A->B->C                                  | 57.03       | 36.67      | 19.52       | 14.47      | 0.23                    | 37.52***         | 0.000005 | 0.000006 | 1.35             |
| A<—>B<-C                                 | 74.07       | 39.69      | 13.34       | 8.54       | 0.41*                   | 60.72***         | 0.000000 | 0.000000 | 2.12             |
| A<—>B->C                                 | 131.86      | 57.91      | 16.72       | 12.01      | 0.61***                 | 115.14***        | 0.000000 | 0.000000 | 2.75             |
| A->B<-C, A->C                            | 35.31       | 24.84      | 7.07        | 7.22       | 0.24                    | 28.24***         | 0.000001 | 0.000001 | 1.54             |
| A<-B<-C, A->C                            | 0.97        | 1.21       | 0.21        | 0.49       | −0.29                   | 0.76**           | 0.008005 | 0.008281 | 0.82             |
| A<—>B<—>C                                | 39.31       | 23.51      | 4.00        | 3.51       | 0.20                    | 35.31***         | 0.000000 | 0.000000 | 2.10             |
| A<-B->C, A<—>C                           | 30.79       | 20.51      | 2.66        | 2.96       | 0.54**                  | 28.14***         | 0.000000 | 0.000000 | 1.92             |
| A->B<-C, A<—>C                           | 27.97       | 13.85      | 3.72        | 2.33       | 0.23                    | 24.24***         | 0.000000 | 0.000000 | 2.44             |
| A->B->C, A<—>C                           | 11.79       | 7.11       | 1.10        | 1.29       | 0.27                    | 10.69***         | 0.000000 | 0.000000 | 2.09             |
| A->B<—>C, A<—>C                          | 42.14       | 19.91      | 2.31        | 2.61       | 0.43*                   | 39.83***         | 0.000000 | 0.000000 | 2.80             |
| A<—>B<—>C, A<—>C                         | 22.21       | 15.56      | 0.79        | 1.08       | 0.54**                  | 21.41***         | 0.000000 | 0.000000 | 1.94             |
| Group-level Indicators                   |             |            |             |            |                         |                  |          |          |                  |
| Density <sup>h</sup>                     | 0.21        | 0.06       | 0.07        | 0.03       | 0.74***                 | 0.14***          | 0.000000 | 0.000000 | 3.16             |
| At least one In-degree <sup>i</sup>      | 26.97(0.98) | 4.06(0.03) | 21.41(0.78) | 4.56(0.12) | 0.72***                 | 5.55***          | 0.000000 | 0.000000 | 1.29             |
| At least one Out-degree <sup>j</sup>     | 25.79(0.93) | 4.24(0.06) | 18.14(0.66) | 4.49(0.13) | 0.72***                 | 7.66***          | 0.000000 | 0.000000 | 1.75             |
| Average relationship number <sup>k</sup> | 5.65        | 1.22       | 1.85        | 0.58       | 0.58***                 | 3.80***          | 0.000000 | 0.000000 | 3.98             |

Table 2 shows the indicators and comparisons of the two types of networks, with the numbers in parentheses indicating the proportion of students with in-degree/out-degree among all students in the class. <sup>a</sup> In-degree refers to the quantity of relationships received by a student in the network. In the FNs, this indicates how many classmates have chosen the student as a friend, and in the SSNs, it indicates how many classmates have sought support from the student. <sup>b</sup> Out-degree refers to the quantity of relationships initiated by a student in the network. In the FNs, this indicates how many classmates the student has chosen as friends, and in the SSNs, it indicates how many classmates the student has sought support from. Among the 799 students, 81 (accounting for 10.14%) had a friendship out-degree exceeding 10, 17 (2.13%) exceeded 15, and 9 (1.13%) exceeded 20. In contrast, the proportions of students with a support out-degree exceeding 10, 15, and 20 were 0.3%, 0.2%, and 0.1%, respectively. <sup>c</sup> Non-reciprocal relationship refers to the number of unidirectional connections between pairs. <sup>d</sup> Reciprocal relationship refers to the number of pairs with mutual connections. <sup>e</sup> Nonexistent refers to the number of pairs without connections. <sup>f</sup> Same gender refers to the proportion of same-sex connections, Boy → boy, Boy → girl, Girl → girl, Girl → boy also refer to the proportion of corresponding connections. <sup>g</sup> Triadic indicators represent the number of 16 possible types of relationships among three nodes in the network, where A, B, and C represent three nodes, and → indicates the connection and direction between two nodes. <sup>h</sup> Density is the observed number of connections divided by the total possible number of connections. <sup>i</sup> At least one in-degree refers to the number of students who have at least one incoming relationship. <sup>j</sup> At least one out-degree refers to the number of students who have at least one outgoing relationship. <sup>k</sup> Average degree refers to the average number of relationships per person in the network. \**p* ≤ 0.05; \*\**p* ≤ 0.01; \*\*\**p* ≤ 0.001, and the same below

difference-test results, there are distinct differences between the two networks at the individual, dyadic, triadic, and group levels, as evidenced by the following observations: At the individual level, the average out-degree and in-degree

of students in their FNs are higher than those observed in SSNs. At the dyadic level, the numbers of non-reciprocal and reciprocal relationships in the FNs exceed those in the SSNs, while the number of unrelated pairs in the FNs is

smaller than in the SSNs. At the triadic level, the number of empty triads in the FNs is approximately half the number observed in the SSNs, and the numbers of various types of connected triads in the FNs consistently exceed those in the SSNs. At the group level, the average density of FNs is three times that of SSNs. The average number of individuals who send out at least one connection is greater in the FNs, as is the average number of individuals who receive at least one connection. Similarly, the average number of relationships per person in the FNs within a class is higher than in SSNs. Additionally, the “Correlation Coefficient” column in Table 2 indicates a moderate-to-high correlation between the structural indicators of the two networks. Therefore, H2 is confirmed.

### Comparison of configuration effects and the role of perceived peer support between the two networks

In this study, over a dozen configurations have been tested, and the results showed that only *Arc*, *Reciprocity*, *Popularity*, and *Activity* consistently provided a good fit for ERGMs across most class networks. Therefore, these four configurations, along with *Sender* and *Receiver* effects, were included as effects to fit both the FN and SSN for each class. The GOF assessment indicated that the fitted ERGMs provided an acceptable representation of the observed class networks, particularly for the structural effects included in the models. The *t*-ratio statistics for the six model terms (*Arcs*, *Reciprocity*, *Popularity*, *Activity*, *Sender*, and *Receiver* effects) were predominantly close to zero across the classes. No class showed poor fit ( $|t| \geq 2$ ) for any effect, and the majority of classes exhibited good fit ( $|t| < 0.5$ ) on all terms. This indicates that the model accurately reproduced the targeted network effects. In contrast, the multivariate GOF statistics – the Mahalanobis distance and the maximum quasi-autocorrelation – showed greater variability and, in some classes, notably high values. This suggests that while the model captures the targeted effects well, it may not fully reproduce all aspects of the network data simultaneously. We therefore interpret the ERGM results as capturing the main structural tendencies of the networks rather than providing a perfect, exhaustive fit.

The “Results of ERGMs” part in Table 3 presents the meta-analysis results of the ERGMs. The section on **Configuration Models** provides a meta-analysis of configuration models across all classes. Since the models of three classes did not converge, they were excluded from the meta-analysis. The *t*-ratios of each local configuration in the two types of networks for the remaining 26 classes are uniformly below 0.1, indicating that the models have successfully converged. The results indicate that the estimates of the four local configuration effects are statistically

significant. The effect size of *Arcs* is negative, reflecting the sparsity of the network. The higher the *Reciprocity* effect, the more students select each other, with an odds ratio of  $e^{2.24} = 9.39$  in FNs and  $e^{2.49} = 12.06$  in SSNs. The *Popularity* effect exerts a positive influence in FNs ( $e^{0.26} = 1.30$ ) and SSNs ( $e^{0.17} = 1.19$ ). Similarly, the *Activity* effect also plays a significant role in FNs ( $e^{1.17} = 3.22$ ) and SSNs ( $e^{0.84} = 2.32$ ). The section on **Actor Effect Models** of the “Results of ERGMs” part in Table 3 presents the meta-analysis results of **Actor Effects** after controlling for **Configuration Effects**. Since students from two classes did not complete the items on perceived peer support, and the models from two other classes did not converge, the meta-analysis included data from 25 classes. The results reveal that the *Sender Effect* has a significant positive effect in both types of networks ( $e^{0.17} = 1.19$ ,  $e^{0.26} = 1.30$ ), while the *Receiver Effect* is not significant.

“The Difference Test” part of Table 3 presents a comparison of the ERGMs’ effect values between the two networks in all classes. The first four rows compare the configuration models, showing that the *Activity* effect in FNs exceeds that of SSNs by correction of FDR. The last two rows of “The Difference Test” part in Table 3 compare the *Sender* and *Receiver* effects between the two networks. Although the *Sender Effect* in SSNs is only marginally higher than that in FNs after the correction of FDR, the Cohen’s *d* is  $-0.5685$ , which represents a medium effect size according to conventional benchmarks (e.g., Cohen, 1988). This indicates that the observed difference, while statistically marginal, is not trivial in magnitude and may hold practical relevance. Considering that the comparison was based on only 25 classes, the limited number of groups likely reduced the statistical power, which may have contributed to the lack of significance. Similarly, although the *t*-tests revealed no statistically significant differences in *Popularity* and *Receiver* effect between the two networks, the observed effect sizes (Cohen’s  $d = -0.25$  and  $-0.37$ , respectively) are non-negligible and fall within the small-to-medium range. These results suggest that the lack of significance may be due to limited statistical power rather than the absence of meaningful differences, highlighting these constructs as candidates for further investigation with larger samples. Overall, the above results partially support H3 and H4.

### Relationship between friendship/support-seeking networks and mental health

Table 4 presents the correlations between the in-degree and out-degree of FNs and SSNs with mental health indicators. Figure 2 illustrates the relationships of in-degree and out-degree with students’ mental health after FDR correction. The results indicate that the out-degrees of both FNs and

**Table 3** The results of ERGMs and the difference tests of local configurations between networks

| Configuration          | Results of ERGMs |      |      |         | The Difference Test |      |       |      |                    |       |       |         |           |
|------------------------|------------------|------|------|---------|---------------------|------|-------|------|--------------------|-------|-------|---------|-----------|
|                        | FNs              |      | SSNs |         | FNs                 |      | SSNs  |      | FNs-SSNs           |       | p     | q       | Cohen's d |
|                        | Est              | SE   | n    | SE      | M                   | SD   | M     | SD   | Value              |       |       |         |           |
| Configuration Models   |                  |      |      |         |                     |      |       |      |                    |       |       |         |           |
| Configuration Effects  |                  |      |      |         |                     |      |       |      |                    |       |       |         |           |
| <i>Arcs</i>            | -4.60***         | 0.13 | 26   | 0.11    | -3.02               | 6.07 | -4.05 | 0.69 | 0.91               | 0.369 | 0.492 | 0.2384  |           |
| <i>Reciprocity</i>     | 2.24***          | 0.11 | 26   | 0.13    | 2.19                | 0.61 | 2.17  | 1.53 | 0.06               | 0.957 | 0.957 | 0.0172  |           |
| <i>Popularity</i>      | 0.26**           | 0.08 | 26   | 0.17*   | -0.58               | 3.24 | -0.00 | 0.53 | -0.96              | 0.348 | 0.492 | -0.2498 |           |
| <i>Activity</i>        | 1.17***          | 0.09 | 26   | 0.84*** | 1.17                | 0.45 | 0.79  | 0.43 | 4.42***            | 0.000 | 0.000 | 0.8634  |           |
| Actor Effect Models    |                  |      |      |         |                     |      |       |      |                    |       |       |         |           |
| Configuration Effects  |                  |      |      |         |                     |      |       |      |                    |       |       |         |           |
| <i>Arcs</i>            | -5.07***         | 0.20 | 25   | 0.24    | -3.99               | 3.02 | -5.43 | 1.62 | 1.44**             | 0.007 | 0.021 | 0.5942  |           |
| <i>Reciprocity</i>     | 2.23***          | 0.12 | 25   | 0.13    | 2.19                | 0.64 | 2.17  | 1.52 | 0.06               | 0.832 | 0.832 | 0.0171  |           |
| <i>Popularity</i>      | 0.35**           | 0.09 | 25   | 0.24**  | -0.23               | 1.62 | 0.07  | 0.58 | -0.30              | 0.283 | 0.340 | -0.2466 |           |
| <i>Activity</i>        | 0.99***          | 0.09 | 25   | 0.71*** | 0.98                | 0.45 | 0.61  | 0.52 | 0.37**             | 0.002 | 0.012 | 0.7609  |           |
| Actor relation effects |                  |      |      |         |                     |      |       |      |                    |       |       |         |           |
| <i>Sender Effect</i>   | 0.17***          | 0.04 | 25   | 0.26*** | 0.22                | 0.22 | 0.40  | 0.39 | -0.18*             | 0.045 | 0.090 | -0.5685 |           |
| <i>Receiver Effect</i> | -0.02            | 0.04 | 25   | 0.01    | -0.03               | 0.20 | 0.07  | 0.32 | -0.10 <sup>+</sup> | 0.088 | 0.132 | -0.3748 |           |

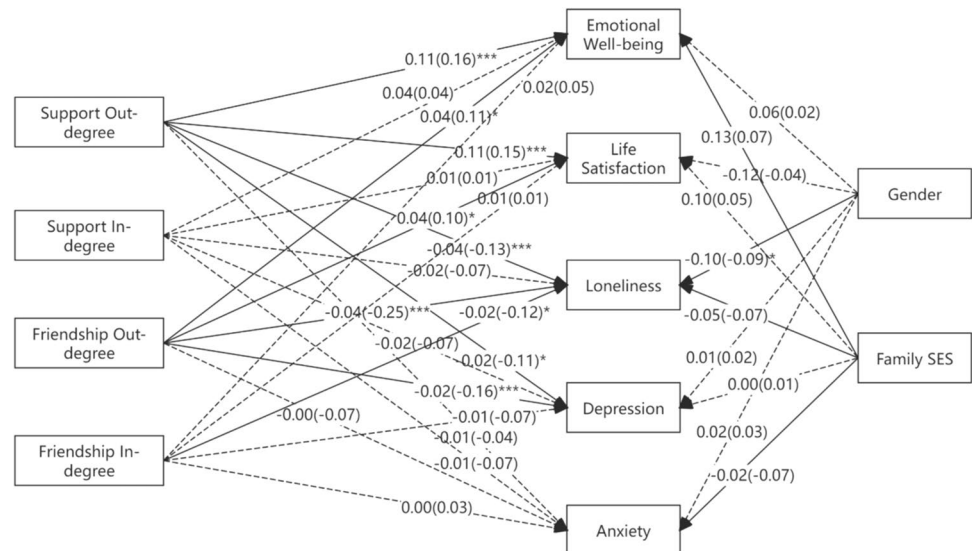
1) *Est.* represents the average estimated effect size calculated in the meta-analysis. 2) *SE* stands for the standard error of the overall estimate. 3) *n* refers to the number of classes included in the meta-analysis

**Table 4** The relationship between in-degree/out-degree and mental health indicators<sup>a</sup>

|                          | 1        | 2        | 3        | 4        | 5        | 6        | 7        | 8        | 9        | 10    | 11                 | 12    |
|--------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|-------|--------------------|-------|
| 1 Support in-degree      | 1        |          |          |          |          |          |          |          |          |       |                    |       |
| 2 Support out-degree     | 0.24***  | 1        |          |          |          |          |          |          |          |       |                    |       |
| 3 Friendship in-degree   | 0.57***  | 0.23***  | 1        |          |          |          |          |          |          |       |                    |       |
| 4 Friendship out-degree  | 0.20***  | 0.39***  | 0.27***  | 1        |          |          |          |          |          |       |                    |       |
| 5 Emotional well-being   | 0.14***  | 0.23***  | 0.14***  | 0.20***  | 1        |          |          |          |          |       |                    |       |
| 6 Life satisfaction      | 0.06     | 0.18***  | 0.07     | 0.15***  | 0.75***  | 1        |          |          |          |       |                    |       |
| 7 Loneliness             | -0.24*** | -0.28*** | -0.26*** | -0.34*** | -0.50*** | -0.44*** | 1        |          |          |       |                    |       |
| 8 Depression             | -0.17*** | -0.20*** | -0.18*** | -0.23*** | -0.63*** | -0.54*** | 0.55***  | 1        |          |       |                    |       |
| 9 Anxiety                | -0.08*   | -0.08*   | -0.05    | -0.09*   | -0.44*** | -0.40*** | 0.41***  | 0.64***  | 1        |       |                    |       |
| 10 Gender                | 0.17***  | 0.07*    | -0.04    | -0.04    | 0.03     | -0.04    | -0.10**  | 0.00     | 0.01     | 1     |                    |       |
| 11 Family SES            | 0.07*    | 0.03     | 0.10**   | 0.03     | 0.09*    | 0.05     | -0.10**  | -0.03    | -0.09*   | -0.01 | 1                  |       |
| 12 Peer Support          | 0.22***  | 0.33***  | 0.18***  | 0.25***  | 0.39***  | 0.31***  | -0.54*** | -0.38*** | -0.19*** | 0.07  | 0.05               | 1     |
| <i>M</i>                 | 1.84     | 1.84     | 5.52     | 5.52     | 5.06     | 5.28     | 0.75     | 0.48     | 0.40     | 1.49  | 0.01               | 2.96  |
| <i>SD</i>                | 1.62     | 2.03     | 2.79     | 4.05     | 1.43     | 1.53     | 0.56     | 0.39     | 0.26     | 0.50  | 0.81               | 0.90  |
| <i>M<sub>boy</sub></i>   | 1.57     | 1.70     | 5.63     | 5.68     | 5.01     | 5.34     | 0.81     | 0.48     | 0.39     | —     | 0.01               | 2.90  |
| <i>SD<sub>boy</sub></i>  | 1.53     | 2.18     | 3.03     | 4.20     | 1.49     | 1.58     | 0.61     | 0.40     | 0.27     | —     | 0.79               | 0.93  |
| <i>M<sub>girl</sub></i>  | 2.13     | 1.99     | 5.41     | 5.36     | 5.11     | 5.21     | 0.69     | 0.48     | 0.40     | —     | 0.00               | 3.02  |
| <i>SD<sub>girl</sub></i> | 1.65     | 1.85     | 2.51     | 3.88     | 1.36     | 1.47     | 0.50     | 0.38     | 0.24     | —     | 0.84               | 0.86  |
| <i>Mdn</i>               | 2.00     | 1.00     | 5.00     | 5.00     | 5.13     | 5.00     | 0.63     | 0.40     | 0.39     | 1.00  | -0.17              | 3.00  |
| <i>Mo</i>                | 1.00     | 0.00     | 5.00     | 3.00     | 7.00     | 7.00     | 0.19     | 0.20     | 0.50     | 1.00  | -0.51 <sup>b</sup> | 4.00  |
| <i>Min</i>               | 0.00     | 0.00     | 0.00     | 0.00     | 1.00     | 1.00     | 0.00     | 0.00     | 0.00     | 1.00  | -1.70              | 0.00  |
| <i>Max</i>               | 9.00     | 21.00    | 16.00    | 26.00    | 7.00     | 7.00     | 3.00     | 2.00     | 1.00     | 2.00  | 2.19               | 4.00  |
| <i>Kurt</i>              | 0.99     | 13.43    | 0.26     | 3.46     | -0.02    | 0.19     | 0.98     | 0.92     | -0.76    | -2.0  | -0.34              | 0.46  |
| <i>Skew</i>              | 1.03     | 2.35     | 0.31     | 1.46     | -0.58    | -0.79    | 0.93     | 0.99     | 0.33     | 0.06  | 0.60               | -0.83 |

<sup>a</sup> In the first column, *M* represents the mean, calculated as the average of each student's mean score on the relevant dimension; *SD* is the standard deviation; *Mdn* is the median; *Mo* is the mode; *Min* is the minimum value; *Max* is the maximum value; *Skew* is skewness; *Kurt* is kurtosis. <sup>b</sup> There are multiple modes in the group. Among these, the smallest mode is shown here

**Fig. 2** Associations of FNs/SSNs with adolescent mental health (799 students). Note: 1. This is a saturated model with RMSEA=0.000, CFI=1.000, TLI=1.000, and SRMR=0.000. 2. The numbers in parentheses are the standardized coefficients, and those outside the parentheses are the unstandardized coefficients



SSNs are positively related to emotional well-being and life satisfaction and negatively related to depression and loneliness. The friendship in-degree is negatively associated with loneliness, whereas the support in-degree is not significantly associated with any mental health indicators.

The Wald tests for the path analysis indicated that there was no significant difference between the two types of out-degree in their relationships with depression, anxiety, and loneliness ( $\chi^2(1)=0.002$ ,  $p=0.9614$ ;  $\chi^2(1)=0.345$ ,  $p=0.5572$ ;  $\chi^2(1)=0.029$ ,  $p=0.8651$ ; respectively). Similarly, no significant difference was found between the two types of in-degrees in their relationships with the mental health indicators (emotional well-being:  $\chi^2(1)=0.061$ ,  $p=0.8052$ ; life satisfaction:  $\chi^2(1)=0.010$ ,  $p=0.9184$ ; loneliness:  $\chi^2(1)=0.002$ ,  $p=0.9643$ ; depression:  $\chi^2(1)=0.309$ ,  $p=0.5780$ ; anxiety:  $\chi^2(1)=1.815$ ,  $p=0.1779$ ; respectively). Furthermore, although the initial analysis suggested that support-seeking out-degree had a stronger relationship with emotional well-being and life satisfaction than friendship out-degree (emotional well-being:  $\chi^2(1)=3.923$ ,  $p=0.0476$ ,  $q=0.2380$ ; life satisfaction:  $\chi^2(1)=4.034$ ,  $p=0.0446$ ,  $q=0.2380$ ), the results of correction of FDR indicated that these differences might not be statistically significant. Therefore, a two-stage testing procedure was employed (Benjamini & Hochberg, 1995; Storey & Tibshirani, 2003). First, comparisons with  $p < 0.05$  in the Wald tests were identified, specifically the comparisons between the two types of out-degree on emotional well-being and life satisfaction. Second, separate regression models were constructed for each outcome variable (emotional well-being and life satisfaction), each including both types of out-degree as predictors. The differences between the corresponding

regression coefficients for the two out-degree were then compared, followed by correction of FDR. The results showed a significant difference for emotional well-being ( $\beta_{\text{support out-degree}}=0.119^{***}$ ,  $\beta_{\text{friendship out-degree}}=0.046^{***}$ , Wald test value=2.5113,  $p=0.0122$ ,  $q=0.0237$ ). The analysis for life satisfaction followed the same procedure ( $\beta_{\text{support out-degree}}=0.108^{***}$ ,  $\beta_{\text{friendship out-degree}}=0.034^*$ , Wald test value=2.2665,  $p=0.0237$ ,  $q=0.0237$ ). These results partially support H5 and H6.

To quantify the unique contribution of different centrality indices in SSNs and FNs to the psychosocial outcomes, we calculated the effect sizes ( $\Delta R^2$ ) through a series of nested model comparisons. As shown in Table 5, the “Full Model” explained 7.6%, 5.0%, 18.3%, 8.4%, and 2.0% of the variance in the five outcome variables (Emotional Well-being, Life Satisfaction, Loneliness, Depression, and Anxiety), respectively. By sequentially constraining each path coefficient to zero, we calculated the  $\Delta R^2$  for each centrality metric and its percentage of the variance explained by the Full Model ( $\Delta R^2\%$ ). The analysis revealed that friendship out-degree was an influential predictor across outcomes. Its unique contribution accounted for a substantial proportion of the explained variance in all five outcomes ( $\Delta R^2\%$  ranging from 11.84% to 27.87%), with the largest share for Loneliness ( $\Delta R^2=.051$ , constituting 27.87% of the variance explained by the full model). In contrast, support out-degree showed the most pronounced unique contribution to Emotional Well-being and Life Satisfaction ( $\Delta R^2\%=25.00\%$  and  $36.00\%$ , respectively). The in-degree in both networks demonstrated relatively minor unique contributions across all outcomes (all  $\Delta R^2\% < 5\%$ ), indicating that receiving ties had limited independent predictive power after controlling for other variables.

**Table 5** Effect sizes of the path analysis model

| Indices                                            | Emotional well-being | Life satisfaction | Loneliness | Depression | Anxiety            |
|----------------------------------------------------|----------------------|-------------------|------------|------------|--------------------|
| Full Model                                         | 0.076***             | 0.050**           | 0.183***   | 0.084***   | 0.020 <sup>+</sup> |
| Model A: Support In-degree=0                       | 0.075***             | 0.050**           | 0.180***   | 0.081***   | 0.017 <sup>+</sup> |
| $\Delta R^2_A$                                     | 0.001                | 0.000             | 0.003      | 0.003      | 0.003              |
| Proportion of variance explained by $\Delta R^2_A$ | 1.32%                | 0.00%             | 1.64%      | 3.57%      | 15.00%             |
| Model B: Support Out-degree=0                      | 0.057***             | 0.032*            | 0.169***   | 0.075***   | 0.018 <sup>+</sup> |
| $\Delta R^2_B$                                     | 0.019                | 0.018             | 0.014      | 0.009      | 0.002              |
| Proportion of variance explained by $\Delta R^2_B$ | 25.00%               | 36.00%            | 7.65%      | 10.71%     | 10.00%             |
| Model C: Friendship In-degree=0                    | 0.075***             | 0.050**           | 0.174***   | 0.081***   | 0.019 <sup>+</sup> |
| $\Delta R^2_C$                                     | 0.001                | 0.000             | 0.009      | 0.003      | 0.001              |
| Proportion of variance explained by $\Delta R^2_C$ | 1.32%                | 0.00%             | 4.92%      | 3.57%      | 5.00%              |
| Model D: Friendship Out-degree=0                   | 0.066***             | 0.042**           | 0.132***   | 0.063***   | 0.016 <sup>+</sup> |
| $\Delta R^2_D$                                     | 0.009                | 0.008             | 0.051      | 0.021      | 0.004              |
| Proportion of variance explained by $\Delta R^2_D$ | 11.84%               | 16.00%            | 27.87%     | 25.00%     | 20.00%             |

## Discussion

This study analyzed the relationships and disparities between FNs and SSNs, revealing that these two networks exhibit both structural distinctions and overlaps. Additionally, although the primary configurations of both networks are similar, their effects differ, with perceived peer support exerting a more substantial impact on the SSNs. Both networks are significantly associated with mental health, with SSNs showing a stronger association with well-being.

## Structural similarities and differences between friendship networks and support-seeking networks

This study found that the majority of structural indicators of the two networks are moderately correlated, indicating a substantial degree of similarity between them. First, this similarity arises because students often seek help from friends (Schacter & Margolin, 2019; Shin & Park, 2023), resulting in partial overlap between friendship and support-seeking relationships, as evidenced by the overlap between the two networks and the correlation of structural indicators. Second, both networks consist of positive relationships, and the development of these relationships is shaped by similar factors. For instance, a positive group atmosphere can foster positive relationships among group members (Khalfaoui et al., 2021), and in classes with a favorable group atmosphere, both types of networks tend to exhibit higher density.

Our analysis reveals a substantial overlap between the two networks, with 88.5% of support-seeking ties coinciding with friendship ties. This high degree of overlap indicates that the SSN is largely built upon the FN, underscoring that students primarily seek support from friends (Schacter & Margolin, 2019) and that friendship serves as a key foundation for support relations (Wang et al., 2023). However, it is crucial to clarify that the SSN is not merely a subset of the FN. The distinction is evident in two key findings. First, approximately 11.5% of support-seeking relationships exist between non-friends. Second, a substantial proportion of friendship ties (roughly 71.5%) are not activated for support-seeking. This pattern suggests a relationship of partial overlap with a strong core-periphery structure: the FN forms the core relational scaffold from which most support-seeking occurs, while the SSN extends beyond this core to include a periphery of functionally specific, non-friend ties. Consequently, while FNs and SSNs are deeply intertwined and can mutually reinforce one another (van Rijsewijk et al., 2020), they are governed by distinct relational logics. The configuration of high overlap without full containment—where the SSN is substantially scaffolded by, yet not confined to, the FN—is precisely the reason why the two networks exhibit both pronounced structural similarities and systematic differences in their overall architecture.

Meanwhile, structural differences between the two networks are evident, as nearly all structural indicators at the individual, dyadic, triadic, and group levels differ, with FNs having more relationships. These differences may be attributed to several factors. First, the conditions for establishing support-seeking relationships are typically more stringent (Barbee & Cunningham, 1995), whereas FNs are composed of relationships that are easier to establish. As a

result, SSNs are generally sparser than FNs, and this is evident at all levels of the network. Second, students do not restrict their support-seeking to friends; for instance, they may seek assistance from classmates who are skilled at problem-solving but are not necessarily familiar (Zander et al., 2019). Consequently, some dyads with support relationships are not connected in the FNs, further contributing to the differences between the two networks. In this study, this difference is reflected in 28.5% of friendship relationships overlapping with 88.5% of support relationships, indicating that approximately two-thirds of friendship relationships do not align with support relationships, and roughly 11.5% of support relationships do not coincide with friendship relationships. It is clear that the differences in the nature of dyadic relationships between the two networks contribute to structural disparities to some degree.

### Similarities and differences between configuration models of the two networks

The results of the configuration models confirm the similarities in local structures between the two networks. FNs and SSNs share overlapping categories of local configurations, namely *Reciprocity*, *Popularity*, and *Activity*. In FNs, students are inclined to choose each other as friends (*Reciprocity*), some students are selected as friends by many classmates (*Popularity*), and some students are motivated to make friends (*Activity*), consistent with previous research findings (Wegemer, 2022). In SSNs, based on principles of fairness and familiarity, students who are sought for help also tend to seek help from peers who have requested their assistance, reflecting *Reciprocity* (Wang et al., 2021); Seeking support from others benefits students, prompting them to seek help from more classmates, thereby promoting *Activity* (Barbee & Cunningham, 1995; Wang et al., 2021); Additionally, students who are frequently sought for help by classmates are typically willing and able to help others (Snippe et al., 2018; Zander et al., 2019), thereby increasing their *popularity*. Consequently, these three local configurations are crucial in both types of networks. The overlap in local configurations strengthens the overall similarity between the two networks.

By comparing the configuration models of each network, it becomes clear that the role of configurations differs between networks. First, *Reciprocity* in the SSNs is higher than in the FNs, indicating that students are more likely to seek help from peers who also seek help from them, with *Reciprocity* playing a greater role in the SSNs. That is, in the class, *mutuality within support ties is higher than mutuality within friendship nominations*. This may be because, when students seek support, they typically turn to close friends, and support relationships with close friends

are often mutual, reflected in higher reciprocity (Rubin et al., 2011; Zander et al., 2019). This reflects the structural tendency of support networks to be more mirrored or reciprocal, similar to networks based on behaviors like "who do you spend the most time with" where mutuality is inherently more common. Second, *Activity* in the SSNs is significantly lower than in the FNs, indicating that, compared to support relationships, friendships tend to be more concentrated among certain students. This may be because the criteria for selecting support-seeking targets are more stringent (Barbee & Cunningham, 1995), and students have fewer peers they can turn to for support, while there are fewer restrictions on forming friendships. As a result, *students are more likely to initiate additional friendships*.

### The differential impact of perceived peer support on network development

This study found that *the Sender effect of perceived peer support is significant in the SSNs*. This phenomenon may arise because *individuals with high levels of perceived support are more inclined to believe that others are willing to assist them*, are more attuned to others' willingness to provide support, and, consequently, are more likely to seek help (Gao et al., 2020). Although this effect is also evident in FNs, its magnitude is significantly lower than in SSNs. Therefore, perceived peer support contributes to structural differences between these two networks to a certain extent.

### Comparison of the relationships of the two networks with mental health

This study found that both FNs and SSNs are closely associated with students' mental health. Students with higher friendship out-degree are more likely to exhibit greater emotional well-being, higher life satisfaction, and lower levels of depression and loneliness. Additionally, students with higher friendship in-degree tend to experience reduced levels of loneliness. Similarly, students with higher support out-degree also report greater emotional well-being, enhanced life satisfaction, and lower levels of depression and loneliness.

Friendship ties fulfill psychological needs and contribute to subjective well-being (Telecan et al., 2024). Friendships provide companionship and intimacy, fostering feelings of respect and connectedness while promoting positive emotions and satisfaction (Lei, 2021). Furthermore, friendships enable emotional expression and regulation, helping students mitigate negative emotions (Yang et al., 2020). A higher friendship out-degree and in-degree imply having a larger number of friends, which reduces feelings of loneliness.

Similarly, support-seeking relationships help students address daily challenges, thereby reducing depression and

loneliness while enhancing overall well-being. Students often face various difficulties. **Having more support-seeking relationships increases the likelihood of obtaining the support that students need to resolve problems, reduce emotional distress, and promote happiness.**

**The relationship of support out-degree with well-being surpasses that of friendship out-degree.** This disparity may arise because actively seeking support enables students to better manage daily challenges and hardships, fostering personal growth and a sense of accomplishment (Fullerton et al., 2021; Letkiewicz et al., 2023). While friendships play a significant role in providing emotional support, they are less directly involved in resolving specific difficulties, thereby exerting a comparatively weaker association with well-being. Moreover, overlaps between SSNs and FNs may represent particularly important relationships (Barbee & Cunningham, 1995). Overall, support out-degree demonstrates a more substantial relationship with well-being.

The effect size analysis ( $\Delta R^2\%$ ) provides a precise quantification of the relative importance of different network features. First, friendship out-degree emerged as the primary factor predicting Loneliness and Depression, with its nearly 28% contribution suggesting that interventions targeting friendship may be particularly potent in alleviating negative psychological states. Second, support out-degree was more strongly linked to positive experiences (Emotional Well-being, Life Satisfaction), accounting for over one-third of the explained variance. This supports the traditional view that active social embeddedness is a cornerstone of well-being. Finally, the minimal independent contribution of “in-degree” (passively received ties), regardless of network type, strongly implies that the psychological benefits of social connectedness are primarily derived from active, initiating behaviors rather than from passively held positions. These findings, underscored by the disparity in effect sizes, carry clear intervention implications: enhancing mental health may depend more on encouraging active, functional social engagement than merely expanding passive social network size.

## Limitations and future directions

This study is subject to several limitations.

First, while it examines the impact of perceived peer support on network structure, previous research indicates that individuals who receive more support from others are more likely to perceive higher levels of support (Haber et al., 2007). This finding implies that **network structure may also influence perceived peer support.** Future research should employ longitudinal designs to explore the bidirectional relationship between them (Ripley et al., 2024). Meanwhile, the items measuring perceived peer support referred

to “support from friends”, which may have narrowed the conceptual scope of peer support. Friends represent a subset of peers, and support from friends constitutes only part of peer support. Theoretically, the perceived peer support measured in this study is more closely associated with FNs. Even so, the association between perceived peer support and SSNs remained stronger. A more comprehensive measure of peer support would be expected to show an even stronger association with SSNs. Future research should adopt more inclusive measures of peer support to further validate these findings.

**Second, the study compares the static structures of the two networks. However, given the interconnections between friendship and support-seeking relationships, these networks likely influence one another.** Previous research on friendship and received social support has shown that they **reinforce each other** (van Rijsewijk et al., 2020), while studies on friendship and support-seeking also suggest **interactive effects between these two networks** (Wang et al., 2021, 2023). **Future research should leverage dynamic network analysis to examine the interactions between these networks.**

Third, support networks are associated with adolescent mental health across multiple levels. For instance, network indicators at both the individual and class levels have distinct effects on mental health (Wang et al., 2026), and such multilevel effects may differ between FNs and SSNs. However, as this study included only 29 class networks, the limited number of classes made it difficult to obtain reliable estimates (Zhang & Wang, 2010), thus preventing an exploration of these multilevel differences. Future research should collect data on FNs and SSNs from a larger number of classes to enable multilevel comparisons.

**Fourth, the nomination method employed in this study could be further refined in future research. The nomination method may influence how networks are constructed—for instance, whether the number of nominations is limited and how many are permitted; similarly, the definition of the network boundary** (e.g., limiting to the class or expanding to the grade or school) is a fundamental decision that **shapes which ties are observed.** Additionally, the wording of nominations, such as nominating “close friends”, may construct more supportive friendship networks; and support networks based on perceived support versus those based on sought support may also differ. Future studies should further construct and compare the differences among these networks to help researchers and educational practitioners analyze friendship and peer support more precisely. Meanwhile, the order of nomination questions may have introduced a potential priming effect, such as friendship nominations were presented before support-seeking nominations. Although this effect is likely limited, it cannot be entirely ruled out. Future research should consider counterbalancing the order

of nomination items across samples or using randomized presentation to minimize potential priming effects.

Fifth, a gender homophily effect could not be incorporated into the ERGM specification because models including this term failed to converge, likely due to the extremely strong gender segregation in these class networks. Consequently, the estimated actor-level effects (e.g., for perceived peer support) should be interpreted with caution, as they may partially reflect unmodeled gender homophily. To assess whether the core findings were driven by this unmodeled structure, we conducted exploratory within-gender subgroup analyses. These analyses showed that the pattern of a stronger association between perceived peer support and support-seeking out-degree (compared to friendship out-degree) was consistent within both boy and girl subgroups (correlations: boys—0.18<sup>\*\*</sup>, 0.26<sup>\*\*\*</sup>, 0.24<sup>\*\*\*</sup>, 0.31<sup>\*\*\*</sup>; girls—0.18<sup>\*\*\*</sup>, 0.25<sup>\*\*\*</sup>, 0.18<sup>\*\*\*</sup>, 0.34<sup>\*\*\*</sup>, for friendship in-/out-degree and support-seeking in-/out-degree, respectively). These results provide ancillary, though not definitive, support for the robustness of the ERGM findings. Future research should use larger samples, longitudinal data, and alternative estimation techniques to incorporate homophily effects and further validate the findings of this study.

Sixth, this study did not examine the differences in mental health among students based on their specific positions across the two networks. Future research could explore this by comparing outcomes for students who, for instance, are both friends and support providers (overlapping membership), those who serve in only one role, and those who are outside both networks. Theoretically, students embedded in both networks may benefit from compounded relational resources, potentially leading to better mental health compared to their peers in more peripheral positions. Furthermore, a more nuanced analysis could classify students into even finer categories by incorporating the direction of ties (e.g., active seeking vs. being sought as a supporter) within the networks. Investigating how these distinct network positions are associated with adolescent mental health remains an important and open question for future studies.

In addition, the interpretative scope of our model is bounded by its specification. The parsimonious ERGMs, while successfully capturing the key actor-level effects, likely omits some higher-order structural effects, as suggested by the elevated multivariate goodness-of-fit statistics. This limitation arises from a focus on model stability and cross-class comparability, which necessitated the exclusion of more complex terms (e.g., triadic effects) and may have been compounded by the modest size of the class networks. Therefore, our findings focus on the core mechanisms that were explicitly modeled. They should be interpreted as capturing the dominant tendencies of the network structure, rather than providing an exhaustive representation. Future

research with larger samples or methods enabling the estimation of more elaborate specifications would help to better capture the full complexity of networks.

Moreover, this study did not further examine the characteristics of non-overlapping ties between friendship and support-seeking networks, such as whether they are associated with individual attributes (e.g., gender, SES, or mental health). A more systematic investigation would provide deeper insights into their formation and function. Future research could explore these issues using more targeted designs and analytical approaches to better understand the conditions under which students seek support beyond their friendship ties.

Finally, this study focused exclusively on middle school students from Beijing, which limits the generalizability of the findings. Future studies should include students from diverse educational levels and regions to enhance the applicability of the results.

In sum, the findings suggest that, considering the structural similarities between FNs and SSNs, they may be the related constructs that capture different aspects of peer relationships—one reflecting active support-seeking, the other reflecting broader friendship ties. Furthermore, these two types of networks have distinct relationships with students' well-being. Consequently, future research should carefully select the appropriate network based on the specific research objectives. For example, SSNs may be more suitable for studies focusing on perceived peer support, whereas FNs are better suited for investigations into the structure and properties of broader friendship ties, such as studying the friendship atmosphere within a class.

## Conclusion

The findings of this study reveal both meaningful distinctions and substantial overlap between friendship networks (FNs) and support-seeking networks (SSNs) in class settings. Structurally, the two networks differ across individual, dyadic, triadic, and group-level indicators, yet they also show moderate-to-high correlations, reflecting a considerable degree of relational overlap. While FNs and SSNs share similar local configuration types, the functions of some configurations vary markedly. Perceived peer support maybe differentially shape the two networks, exerting a marginally stronger influence on SSNs than on FNs. Finally, both networks are associated with students' mental health, though SSNs appear more closely linked to well-being. Together, these results underscore the importance of distinguishing between FNs and SSNs in peer relationship research. Although related, they capture functionally distinct aspects of adolescent social life and have differing implications for psychological adjustment.

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**Data availability** The data supporting the findings of this study are available from Longitudinal Study on Social Development of Children and Adolescents (LSSDCA). However, access to these data is restricted as they were used under a license for the current study and are not publicly available. The data may be available from the authors upon reasonable request and with the permission of LSSDCA.

## Declarations

**Ethical approval** This study involving human participants was reviewed and approved by Scientific Research Ethics Committee, Collaborative Innovation Center of Assessment toward Basic Education Quality, Beijing Normal University.

**Consent to participate** Informed consent was obtained from all individual participants included in the study and their parents.

**Consent to publish** All participants provided informed consent for the use of their anonymized data in scientific research. This study analyzed a large-scale database from which all personally identifiable information had been removed prior to analysis. Consequently, only group-level results are presented, with no individual-level data being analyzed or disclosed. Therefore, in accordance with the approved ethical protocol, the requirement for consent to publish individual-level research results was waived for this study.

**Conflict of interest** The authors have no conflicts of interest to report.

## References

- Alsarrani, A., Hunter, R. F., Dunne, L., & Garcia, L. (2022). Association between friendship quality and subjective wellbeing among adolescents: A systematic review. *BMC Public Health*, 22(1), 2420. <https://doi.org/10.1186/s12889-022-14776-4>
- Asher, S. R., Hymel, S., & Renshaw, P. D. (1984). Loneliness in children. *Child Development*, 55(4), 1456–1464. <https://doi.org/10.2307/1130015>
- Barbee, A. P., & Cunningham, M. R. (1995). An experimental approach to social support communications: Interactive coping in close relationships. *Annals of the International Communication Association*, 18(1), 381–413. <https://doi.org/10.1080/23808985.1995.11678921>
- Barrow, E., & Thomas, G. (2022). Exploring perceived barriers and facilitators to mental health help-seeking in adolescents: A systematic literature review. *Educational Psychology in Practice*, 38(2), 173–193. <https://doi.org/10.1080/02667363.2022.2051441>
- Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal statistical society: series B (Methodological)*, 57(1), 289–300. <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>
- Butts, C. T. (2024). Package ‘sna’. Available online at: <https://cran.r-project.org/web/packages/sna>. Accessed 26 Sept 2024.
- Campbell, A., Converse, P. E., & Rodgers, W. L. (1976). *The Quality of American Life: Perceptions, Evaluations, and Satisfaction*. Russell Sage Foundation.
- Cheng, C., & Huang, X. (2021). The influence of individually-oriented courage on middle school students’ academic achievement: The mediating roles of problem-solving and help-seeking coping styles. *Psychological Development and Education*, 37(01), 60–67. <https://doi.org/10.16187/j.cnki.issn1001-4918.2021.01.08>
- Cohen, J. (1988). *Statistical power analysis for the behavioral sciences*. Lawrence Erlbaum Associates Publishers.
- Csárdi, M. G. (2024). Package ‘igraph’. Available online at: <https://cran.r-project.org/web/packages/igraph/igraph.pdf>. Accessed 25 Sept 2024.
- Dong, Q., & Lin, C. (2011). *Psychological Development Indicators and Assessment for Children and Adolescents Aged 6–15 in China*. China Science Publishing & Media Ltd.
- Fullerton, D. J., Zhang, L. M., & Kleitman, S. (2021). An integrative process model of resilience in an academic context: resilience resources, coping strategies, and positive adaptation. *PLoS ONE*, 16(2), e0246000. <https://doi.org/10.1371/journal.pone.0246000>
- Gao, B., Gao, X., Wu, J., & Wang, J. (2020). The relationship between proactive personality and psychological help-seeking intention in college students: A chain mediating model. *Psychological Research*, 13(06), 566–573.
- Haber, M. G., Cohen, J. L., Lucas, T., & Baltes, B. B. (2007). The relationship between self-reported received and perceived social support: A meta-analytic review. *American Journal of Community Psychology*, 39(1–2), 133–144. <https://doi.org/10.1007/s10464-007-9100-9>
- He, B., & Wang, X. (2023). The influence of physical activity motivation on subjective well-being of college students: The mediating effect of perceived social support. *Sichuan Sports Science*, 42(03), 52–54+82. <https://doi.org/10.13932/j.cnki.sctyx.2023.03.11>
- Khalfaooui, A., García-Carrión, R., & Villardón-Gallego, L. (2021). A systematic review of the literature on aspects affecting positive classroom climate in multicultural early childhood education. *Early Childhood Education Journal*, 49(1), 71–81. <https://doi.org/10.1007/s10643-020-01054-4>
- Kovacs, M., & Beck, A. T. (1977). An Empirical-Clinical Approach toward a Definition of Childhood Depression. In J. G. Schulerbrandt & A. Raskin (Eds.), *Depression in Childhood: Diagnosis, Treatment, and Conceptual Models* (pp. 1–25). Raven Press.
- Lee, R. P., Ruan, D., & Lai, G. (2005). Social structure and support networks in Beijing and Hong Kong. *Social Networks*, 27(3), 249–274. <https://doi.org/10.1016/j.socnet.2005.04.001>
- Lei, L. (2021). *Developmental Psychology*. China Renmin University Press.

- Letkiewicz, A. M., Li, L. Y., Hoffman, L. M., & Shankman, S. A. (2023). A prospective study of the relative contribution of adolescent peer support quantity and quality to depressive symptoms. *Journal of Child Psychology and Psychiatry*, 64(9), 1314–1323. <https://doi.org/10.1111/jcpp.13813>
- Li, M., & Stone, H. N. (2018). A social network analysis of the impact of a teacher and student community on academic motivation in a science classroom. *Societies*, 8(3), 68. <https://doi.org/10.3390/soc8030068>
- Li, Z., Zhao, M., Xiao, Z., & Liu, Y. (2018). Altruistic behavior and the meaning in life: The mediating effects of perceived social support and self-esteem. *Chinese Journal of Special Education*, 2018(5), 92–96.
- Li, R., Che Hassan, N., & Saharuddin, N. (2023). College student's academic help-seeking behavior: a systematic literature review. *Behavioral Sciences*, 13(8), 637. <https://doi.org/10.3390/bs13080637>
- Liu, J. (2019). *Whole network approach: A practical guide to UCINET*. Truth & Wisdom Press.
- Lubbers, M. J., & Snijders, T. A. (2007). A comparison of various approaches to the exponential random graph model: A reanalysis of 102 student networks in school classes. *Social Networks*, 29(4), 489–507. <https://doi.org/10.1016/j.socnet.2007.03.002>
- Lusher, D., Koskinen, J., & Robins, G. (Eds.). (2013). *Exponential Random Graph Models for Social Networks: Theory, Methods, and Applications* (Vol. Vol. 35). Cambridge University Press.
- Melkman, E. P. (2017). Childhood adversity, social support networks and well-being among youth aging out of care: An exploratory study of mediation. *Child Abuse & Neglect*, 72, 85–97. <https://doi.org/10.1016/j.chiabu.2017.07.020>
- Niu, K., Li, M., & Zhang, X. (2021). The relationship between adolescents' friendship quality and subjective well-being: A meta-analysis. *Psychological Development and Education*, 37(03), 407–418. <https://doi.org/10.16187/j.cnki.issn1001-4918.2021.03.12>
- OECD. (2012). *PISA 2009 technical report*. OECD Publishing.
- Reynolds, C. R., & Richmond, B. O. (1978). What i think and feel: A revised measure of children's manifest anxiety. *Journal of Abnormal Child Psychology*, 6(2), 271–280. <https://doi.org/10.1007/BF00919131>
- Ripley, R. M., Snijders, T. A., Boda, Z., Vörös, A., & Preciado, P. (2024). *Manual for RSIENA*. Oxford University Press.
- Rubin, K. H., Bukowski, W. M., & Laursen, B. (Eds.). (2011). *Handbook of peer interactions, relationships, and groups*. Guilford Press.
- Santini, Z. I., Koyanagi, A., Tyrovolas, S., Haro, J. M., & Koushede, V. (2019). The association of social support networks and loneliness with negative perceptions of ageing: Evidence from the Irish longitudinal study on ageing (TILDA). *Ageing and Society*, 39(5), 1070–1090. <https://doi.org/10.1017/S0144686X17001465>
- Schacter, H. L., & Margolin, G. (2019). The interplay of friends and parents in adolescents' daily lives: Towards a dynamic view of social support. *Social Development*, 28(3), 708–724. <https://doi.org/10.1111/sode.12363>
- Schwartz-Mette, R. A., Shankman, J., Dueweke, A. R., Borowski, S., & Rose, A. J. (2020). Relations of friendship experiences with depressive symptoms and loneliness in childhood and adolescence: A meta-analytic review. *Psychological Bulletin*, 146(8), 664. <https://doi.org/10.1037/bul0000239>
- Shin, H., & Park, C. (2023). Emerging adults' help seeking from friends: The role of perceived goal structures and implications for adjustment. *Current Psychology*, 42(17), 14569–14583. <https://doi.org/10.1007/s12144-022-02774-z>
- Snippe, E., Jeronimus, B. F., aan het Rot, M., Bos, E. H., de Jonge, P., & Wichers, M. (2018). The reciprocity of prosocial behavior and positive affect in daily life. *Journal of Personality*, 86(2), 139–146.
- Storey, J. D., & Tibshirani, R. (2003). Statistical significance for genomewide studies. *Proceedings of the National Academy of Sciences*, 100(16), 9440–9445. <https://doi.org/10.1073/pnas.1530509100>
- Telecan, M. I., Curseu, P. L., & Rus, C. L. (2024). How many friends at work are too many? The nonlinear association between the number of friends, social support and mental well-being. *Central European Management Journal*, 32(4), 604–617. <https://doi.org/10.1108/CEMJ-09-2023-0372>
- van den Toren, S. J., van Grieken, A., Lugtenberg, M., Boelens, M., & Raat, H. (2020). Adolescents' views on seeking help for emotional and behavioral problems: A focus group study. *International Journal of Environmental Research and Public Health*, 17(1), 191. <https://doi.org/10.3390/ijerph17010191>
- van Rijsewijk, L. G., Snijders, T. A., Dijkstra, J. K., Steglich, C., & Veenstra, R. (2020). The interplay between adolescents' friendships and the exchange of help: A longitudinal multiplex social network study. *Journal of Research on Adolescence*, 30(1), 63–77. <https://doi.org/10.1111/jora.12501>
- Virtanen, I. A., & Isotalus, P. (2011). The essence of social support in interpersonal communication. *Empedocles: European Journal for the Philosophy of Communication*, 3(1), 25–42. [https://doi.org/10.1386/ejpc.3.1.25\\_1](https://doi.org/10.1386/ejpc.3.1.25_1)
- Wan, C., & Meng, Y. (2024). Chinese teenagers mental health research in recent twenty years: Research hot-spots, the mapping knowledge domains and development trends. *Psychological Research*, 17(01), 87–96.
- Wang, L., Liang, L., Liu, Z., Yuan, K., Ju, J., & Bian, Y. (2021). The developmental process of peer support networks: The role of friendship. *Frontiers in Psychology*, 12, 615148. <https://doi.org/10.3389/fpsyg.2021.615148>
- Wang, L., Zhang, H., Ju, J., & Bian, Y. (2023). The influence of similarity in academic performance and friendship on the adolescents' support-seeking relationship: Based on a longitudinal social network analysis. *Chinese Journal of Applied Psychology*, 29(02), 137–146. <https://doi.org/10.20058/j.cnki.cjap.021197>
- Wang, L., Zhang, H., & Bian, Y. (2026). Individual-Level and Group-Level Peer Support and Adolescent Internalizing Problems: Based on Social Network Analysis. *Journal of Psychological Science*, 49(1), 105–120. <https://doi.org/10.16719/j.cnki.1671-6981.20260111>
- Wang, M. T., Degol, J. L., Amemiya, J., Parr, A., & Guo, J. (2020). Classroom climate and children's academic and psychological wellbeing: A systematic review and meta-analysis. *Developmental Review*, 57, 100912. <https://doi.org/10.1016/j.dr.2020.100912>
- Webster, D., Dunne, L., & Hunter, R. (2021). Association between social networks and subjective well-being in adolescents: A systematic review. *Youth & Society*, 53(2), 175–210. <https://doi.org/10.1177/0044118X20919589>
- Wegemer, C. M. (2022). Service, activism, and friendships in high school: A longitudinal social network analysis of peer influence and critical beliefs. *Journal of Youth and Adolescence*, 51(1), 1–15. <https://doi.org/10.1007/s10964-021-01549-2>
- Wu, F., & Sheng, Y. (2019). Social support network, social support, self-efficacy, health-promoting behavior and healthy aging among older adults: A pathway analysis. *Archives of Gerontology and Geriatrics*, 85, 103934. <https://doi.org/10.1016/j.archger.2019.103934>

- Yang, Y., Chen, L., Chen, G., & Zhang, W. (2020). Peer rejection, friendship support and adolescent depressive symptoms: A mediated moderation model. *Chinese Journal of Clinical Psychology*, 28(02), 348–353. <https://doi.org/10.16128/j.cnki.1005-3611.2020.02.027>
- Zander, L., Chen, I. C., & Hannover, B. (2019). Who asks whom for help in mathematics? A sociometric analysis of adolescents' help-seeking within and beyond clique boundaries. *Learning and Individual Differences*, 72, 49–58. <https://doi.org/10.1016/j.lindif.2019.03.002>
- Zhang, X., & Wang, J. (2010). A Study on Sample Size Issues in Hierarchical Linear Modeling. *Statistics & Decision*, (15), 4–8. <https://doi.org/10.13546/j.cnki.tjyjc.2010.15.003>
- Zimet, G. D., Dahlem, N. W., Zimet, S. G., & Farley, G. K. (1988). The multidimensional scale of perceived social support. *Journal of Personality Assessment*, 52(1), 30–41. [https://doi.org/10.1207/s15327752jpa5201\\_2](https://doi.org/10.1207/s15327752jpa5201_2)

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